AGENT EVALUATION
SCIENCE

Evaluating Agent Swarms: Token Budgets, Active Workers, and Baseline Choice

Maximilian Nicholson[1,2] Ayda Yazdani[1]
[1] University of Bath [2] Amazon
mn866@bath.ac.uk ay582@bath.ac.uk

Abstract

Evaluating an agent swarm requires distinguishing the effects of increasing the total token budget, dividing a fixed ceiling, and changing coordination. We examine these comparisons in 8,400 controlled trials on fresh arithmetic and graph-colouring tasks. At 4,096 completion tokens per worker, increasing independent width from one to four improves accuracy by 16.0 percentage points on both families while the total ceiling quadruples. Splitting a fixed 4,096-token completion ceiling instead lowers graph accuracy from 56.0% to 18.5%; the arithmetic change is inconclusive. The primary comparisons do not establish an accuracy advantage over a sequential control matched on scheduled calls and nominal token ceiling. Ring communication provides no consistent gain. Eleven complementary scaling studies add 20,376 held-out trials across seven task families, including a fresh-task sixteen-worker confirmation that does not establish superiority over fixed-four execution. Candidate and request audits show that finding a valid answer after an initial failure depends on the token allowance. They also reveal incomplete worker activation and 97 controlled-study attempts with unknown usage. The findings show how resource controls, output selection, and execution records qualify claims about agent scaling. They support evaluating specific allocation policies; equal realized compute advantages and general scaling laws remain unestablished from this work.

Keywords: Agent evaluation; Multi-agent systems; Test-time compute; Token budgets; Resource allocation; Coordination; Reliability; Reproducibility

> "All happy families are alike; each unhappy family is unhappy in its own way." —Leo Tolstoy, *Anna Karenina* (Tolstoy, 2014).
>
> *Adapted to this study:* All successful swarms share a valid outcome; unsuccessful swarms fail in different ways.

1 Introduction

Adding agents changes more than the number of contributors. Under a shared token ceiling, wider systems receive smaller per-agent allowances, generate more candidates, and may use different amounts of the available budget; where this may be tokens, time or reasoning. A nominal worker can also fail to become active producing waste or repeated reasoning. An observed accuracy difference therefore measures the deployed allocation and selection procedure, not an isolated effect of collaboration.

This paper asks what fixed budget worker-count evaluations can establish about task success, resource efficiency, and communication. We examine 8,400 controlled trials on fresh arithmetic and graph-colouring tasks, alongside eleven complementary studies on programming, logical reasoning, graph colouring, arithmetic search, and planning. Three distinctions organize the analysis: a ceiling versus realized token use; requested versus active workers; and improvement over one worker versus improvement over a strong fixed-width baseline. These distinctions matter even when task grading is deterministic or not.

Our contribution combines controlled allocation and coordination comparisons with an empirical audit of saved configuration grids, including negative results, feasibility diagnostics, and a fresh-task policy confirmation. We provide all 176 held-out cells in the repository, retain repeated tasks as statistical clusters, and make the limits of the token ledger explicit. The resulting reporting recommendations are derived from the audit; they have not themselves been validated as an evaluation standard.

Relation to prior work. Repeated sampling separates solution coverage from successful candidate selection (Brown et al., 2024); self-consistency and debate provide different aggregation and interaction mechanisms (Wang et al., 2023; Du et al., 2023). Effective inference allocation depends on task difficulty (Snell et al., 2024), and controlled agent-system studies already show dependence on architecture and task structure (Kim et al., 2026). We do not claim task dependence as a new principle. Shared-workspace communication can help discovery tasks given sufficient resources and feedback (Park et al., 2026); our single-round ring comparison tests a narrower intervention. Failure taxonomies (Cemri et al., 2025) complement our resource and activation diagnostics, which do not by themselves identify a causal failure mechanism.

2 Evaluation Design

Figure 1 shows the controlled schedules and token allowances. The complementary studies below retain their original allocations and selection protocols.

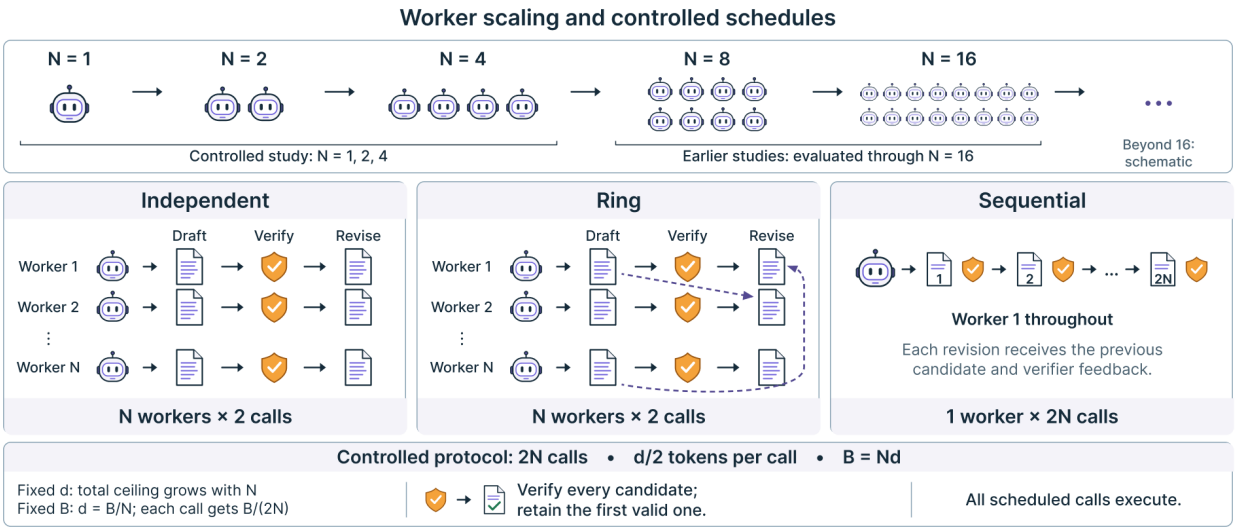


Figure 1 | Worker scaling and controlled schedules. N denotes parallel width, d the per-worker completion allowance, and $B = Nd$ the total completion ceiling. All controlled schedules execute $2N$ calls, each allowed $d/2$ completion tokens. The controlled study tests $N = 1, 2, 4$; earlier studies extend through $N = 16$. Widths beyond sixteen are schematic.

Controlled allocation comparisons. The controlled study comprises 8,400 trials on fresh arithmetic and graph-colouring tasks at widths 1, 2, and 4. Fixing d increases B with width; fixing B reduces each worker's allowance. Independent workers revise their own drafts; ring workers receive predecessor information. The sequential control uses one worker for the same $2N$ scheduled calls and nominal completion ceiling, with prior-candidate and verifier feedback. All scheduled calls execute, and the first valid candidate is retained. The repository contains the supplied protocol summary and states the coverage of the available experimental records.

Evidence and task coverage. Table 1 summarizes the eleven complementary studies. Public task sources are BigCodeBench (Zhuo et al., 2025), MBPP with EvalPlus tests (Austin et al., 2021; Liu et al., 2023), ZebraLogic (Lin et al., 2025), BBEH (Kazemi et al., 2025), and PlanBench (Valmeekam et al., 2023). Graph-colouring and Countdown instances are generated algorithmically and scored by deterministic constraint checks; graph generation is inspired by Reasoning Gym (Stojanovski et al., 2025). These are fixed subsets or adaptations, not full-benchmark leaderboard scores. Luna 5.6, Luna 6, and Muse Contributor are recorded model identifiers; they are not interchangeable model versions (Appendix A).

Task family / study	Models	Studies	Tasks*	Eval. trials
BigCodeBench	L5.6, M	2	32	3,456
MBPP+	L6, M	2	16	864
ZebraLogic original	L6	1	64	3,456
ZebraLogic confirmation	L6	1	300	6,600
Custom graph colouring	L6, M	2	100	2,400
Custom Countdown	L6	1	100	1,200
BBEH subset	L6	1	100	1,200
PlanBench subset	L6	1	100	1,200
Total		11		20,376

Table 1 | Completed evaluation evidence. L5.6/L6: Luna 5.6/6; M: Muse Spark Contributor. *Distinct evaluation tasks per study; task IDs can overlap between model studies. MBPP+ uses one repetition per task and configuration; all others use two. An additional 6,272 calibration trials give 26,648 total. Pilot, smoke, imported baseline, incomplete, and superseded records are excluded.

Interventions and selection. The original programming and ZebraLogic grids evaluate $N \in \{1, 2, 4, 8, 16\}$ at three ceilings. The five smaller diagnostic studies compare one and four workers at two ceilings. The confirmation evaluates a transferred policy and its controls. Workers draft and revise; ring revision adds a predecessor's draft to the revision context. Independent revision omits that peer draft. Both conditions can still share a final integrator in the programming and ZebraLogic workflows. Thus the contrast does not remove every form of information sharing.

Graph colouring, Countdown, and PlanBench select a locally valid revision when available. Their width effects include verifier-assisted parallel search. BBEH instead uses gold-blind plurality with deterministic tie breaking. Programming and ZebraLogic use a model integrator, with final grading outside the generation loop. Appendix A records allocations and graders.

Outcomes and resource accounting. For a complementary study, nominal ceiling B, width N, and revision condition c, let $\widehat{S}_{N,c}(B)$ denote mean final-task success and $\overline{T}_{N,c}(B)$ mean recorded tokens. We compare each wider cell to its shared $N = 1$ control:

$$\Delta S = 100\big(\widehat{S}_{N,c}(B) - \widehat{S}_1(B)\big), \qquad R_T = \frac{\overline{T}_{N,c}(B)}{\overline{T}_1(B)}. \tag{1}$$

ΔS is in percentage points (pp). T retains each study's charged-token or retained-debit convention, including conservative debits where applicable. It is not a complete ledger of all attempted calls or provider compute. Equal ceilings do not imply equal realized resources. Original calibration also contains disclosed accounting overruns; nominal ceilings are not a claim of universally verified hard caps (Appendix B). These charged-token and retained-debit ceilings include prompt accounting; Figure 1 instead specifies a completion-token ceiling for the controlled schedules.

Uncertainty and analysis units. Cells share tasks, repetitions, and controls. Paired bootstrap intervals resample whole tasks, keeping their observations together and using the study-specific strata (Appendix C); detailed resampling settings are recorded in the repository. The 26,648 trials are not independent task samples. Configuration counts describe the evaluated grids; they do not estimate a population frequency or constitute 148 independent hypothesis tests. Reported 95% intervals are study-specific and are not adjusted for cross-study multiplicity.

3 Findings and Their Evaluation Consequences

3.1 Controlled allocation and coordination

The controlled comparisons separate increasing the total allowance from dividing a fixed ceiling (Table 2). Increasing independent width at a fixed per-worker allowance improves accuracy while expanding the total ceiling. Under a fixed total ceiling, graph accuracy falls sharply; the arithmetic comparison is inconclusive. These are comparisons of allocation rules, not evidence of equal-realized-compute gains.

Independent width 1 to 4	Arithmetic	Graph colouring
Fixed per-worker allowance $d = 4{,}096$	$+16.0$ pp	$+16.0$ pp
Fixed total ceiling $B = 4{,}096$	—	-37.5 pp

Table 2 | Controlled-study accuracy changes in percentage points. An em dash indicates an unreported numerical estimate. The first row quadruples the total ceiling.

The primary comparisons do not establish an accuracy advantage over the sequential control matched on scheduled calls and nominal ceiling. Ring communication provides no consistent gain. The candidate audit reports that valid-answer recovery after an initial failure depends on the token allowance. Request records include 97 attempts with unknown usage, limiting exact resource accounting. The repository provides the controlled-study audit summary and states its evidence coverage.

3.2 A shared ceiling does not equalize resource use

All 148 wider held-out cells use more recorded tokens than their matched-ceiling control. Accuracy is lower in 75, tied in 24, and higher in 49 (Figure 2A). Thus 99 cells have higher recorded token use without a higher mean success rate. Programming contributes 96 wider cells, 81 of them flat or worse; the overall count reflects unequal grid sizes and must not be read as a benchmark-balanced effect.

The finding is a joint statement about observed accuracy and the recorded resource ledger. It does not establish population-level dominance for every cell, nor does it compare policies at equal realized expenditure. For example, independent four-worker Countdown gains 9.5–17.0 pp, but uses 2.77–3.85 times the control's recorded tokens. Reporting only accuracy at the shared ceiling would omit that resource difference.

3.3 Width can cross a feasibility boundary

On the PlanBench subset, independent four-worker accuracy is 0% at $B = 8{,}192$, versus 95% for one worker. At $B = 16{,}384$, the same widths score 100% and 89.5%. Their paired differences are -95.0 pp ($[-98.0, -91.5]$) and $+10.5$ pp ($[6.0, 15.5]$). Four-worker length-limit finishes fall from 99.94% to 0.19% of retained stage calls. The higher-ceiling gain uses 4.06 times the control's retained debit.

BBEH provides a complementary case without verifier-assisted selection: ring four-worker accuracy changes from a 6.5-pp loss ($[-11.5, -2.0]$) to a 4.5-pp gain ($[0.5, 9.0]$) when its ceiling dou-

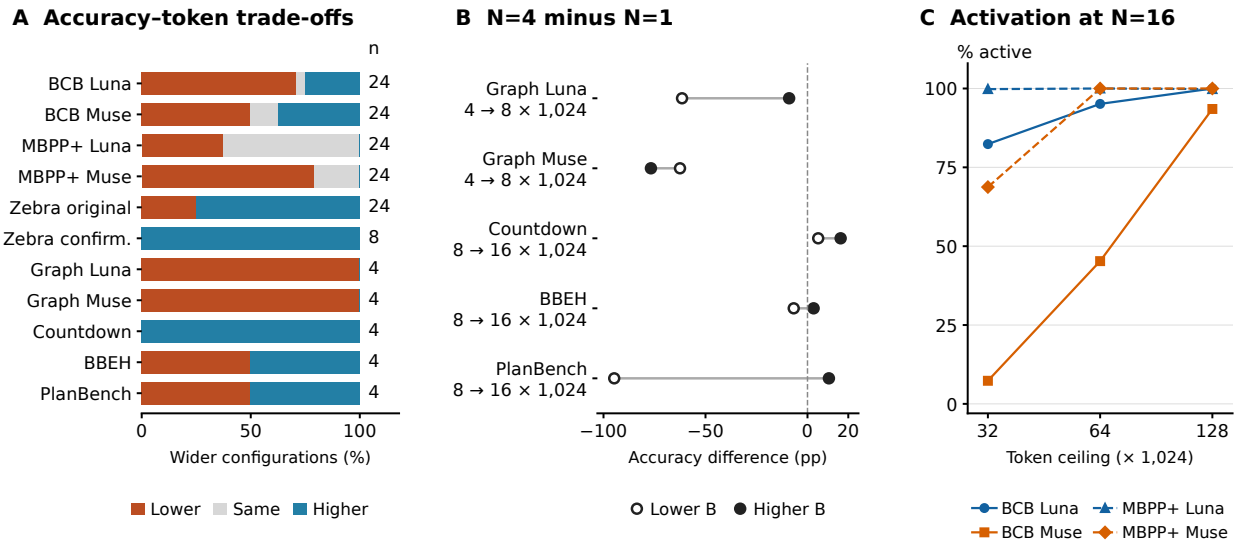


Figure 2 | Three views of the saved evaluation evidence. (A) Fractions of wider cells with lower, equal, or higher accuracy than matched-ceiling $N = 1$; every wider cell uses more recorded tokens ($n = 148$ configurations, not independent samples). (B) Four-worker accuracy effects, pooled over revision conditions, in the five two-ceiling diagnostic studies. (C) Recorded activation, pooled over revision conditions, in the four programming studies that report it. Complete cell values are in the versioned repository; paired communication intervals appear in Appendix C.

bles. Conversely, graph-colouring four-worker cells lose 7.5–77.5 pp across both models and ceilings. Muse's graph-colouring scores include seven operational failure zeros at the higher ceiling (Appendix B). The appropriate interpretation is conditional feasibility under the tested allocation. Width and per-worker allowance change together; lower truncation accompanies the reversals but is not isolated as their cause.

Requested width also differs from recorded participation (Figure 2C). Pooling revision conditions, BigCodeBench Muse's sixteen requested workers average 7.3% activation at the smallest ceiling, rising to 45.3% and 93.5% at the next two ceilings. A worker-count label therefore does not identify a stable amount of realized participation. This diagnostic does not establish that activation alone explains accuracy.

3.4 Sampling gains do not establish communication gains

Countdown improves with independent workers while ring revision reduces average accuracy by 5.0 pp ($[-8.5, -1.75]$). Luna graph colouring also has a negative ring contrast, -2.0 pp ($[-4.0, -0.25]$). Most other communication intervals include or touch zero (Appendix C). PlanBench's saved stratified interval is degenerate because its only differing task occupies a singleton stratum; an unstratified sensitivity interval spans $[0.0, 0.75]$ pp. It is not evidence of general communication superiority.

These contrasts concern the tested ring intervention. Verifier-assisted candidate selection, model integration, and peer exchange are distinct mechanisms. A wider system's gain cannot be assigned to collaboration merely because multiple workers contributed candidates.

3.5 Fresh tasks require a strong baseline as well

The original, post-start-amended ZebraLogic study reports a selected-policy gain of 3.78 pp ($[1.30, 6.51]$) against its calibration-selected fixed-width baseline, which selected one worker. The confirmation transfers the communicating $4/8/16$ policy without refitting to 300 unused puzzles and compares it with communicating fixed four. Averaged over the two higher ceilings, adaptive minus fixed-four accuracy is $+0.92$ pp ($[-0.58, 2.50]$).

The original and confirmation contrasts have different comparators and cannot be pooled.

The confirmation establishes neither superiority nor equivalence. Fresh tasks address task reuse, while a competitive fixed-width control addresses whether adaptation adds value. Both checks are necessary before interpreting a selected allocation rule as an established advantage.

4 Reporting Implications

The audit motivates reporting ceilings and realized use, requested and active workers, per-stage allowances, selection and feedback access, comparator policies, task counts, and paired uncertainty alongside the final score. A structured reporting checklist is provided in the repository. These are practical requirements for interpreting this class of comparison, not an empirically validated new metric.

Further experiments should compare accuracy and realized resource use while reporting width, per-worker allowance, selection, and feedback access. Frozen adaptive policies should then face both single-worker and fixed-width controls on new tasks with matched realized resources. A larger ceiling alone is insufficient to make that comparison.

5 Limitations and Reproducibility

The studies are heterogeneous, use unequal grids, and are not replications from independent laboratories. Programming samples contain only 16–32 evaluation tasks per study. Public-task exposure and unknown training contamination limit generalization. Deterministic grading does not establish that a task subset represents deployed agent work. Most protocols are short draft–revision workflows rather than long-horizon, tool-using environments.

Model versions, provider routes, tier verification, and historical accounting differ. Usage from lost calls and retry expenditure can be unknown. The original ZebraLogic study and confirmation include disclosed operational amendments; the evidence is not uniformly preregistered. Several pilot truncation warnings were acknowledged before the frozen diagnostic grids continued. We retain those failures rather than selecting only feasible cells. No scaling exponent, exact compute comparison, latency advantage, or causal effect of truncation is established.

The appendices retain essential protocols, interval definitions, and interpretation caveats for the eleven complementary studies. Their extended methods, recorded configurations, complete evaluation-cell table, saved outcomes, provenance hashes, and aggregate verifier are in a versioned public repository.[1] The repository also provides the controlled-study protocol and result summary, with an explicit coverage statement for its supporting records. The current materials support saved-summary verification, not complete inference replay or independent grader validation. No new model runs were performed for this reframing.

6 Conclusion

Worker-count comparisons are interpretable only with their allocation, participation, selection, and baseline conditions. The controlled study separates total-budget expansion from division of a fixed allowance and does not establish an advantage over matched sequential execution. The complementary studies show that extra workers can consume more recorded tokens without improving accuracy, budget changes can reverse width effects, and a transferred policy can remain inconclusive against fixed four. These findings support evaluation of specific allocation policies; equal-realized-compute advantages and general scaling laws remain unestablished.

[1] https://github.com/PluvioXO/agent-swarms-reproducibility

References

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, et al. Program synthesis with large language models, 2021. https://arxiv.org/abs/2108.07732. Cited at: 1.

Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, et al. Large language monkeys: Scaling inference compute with repeated sampling, 2024. https://arxiv.org/abs/2407.21787. Cited at: 1.

Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, et al. Why do multi-agent llm systems fail?, 2025. https://arxiv.org/abs/2503.13657. Cited at: 1.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate, 2023. https://arxiv.org/abs/2305.14325. Cited at: 1.

Mehran Kazemi, Bahare Fatemi, Hritik Bansal, John Palowitch, Chrysovalantis Anastasiou, Sanket Vaibhav Mehta, et al. Big-bench extra hard, 2025. https://arxiv.org/abs/2502.19187. Cited at: 1.

Yubin Kim, Ken Gu, Chanwoo Park, Chunjong Park, Samuel Schmidgall, A. Ali Heydari, et al. Towards a science of scaling agent systems, 2026. https://arxiv.org/abs/2512.08296v3. Cited at: 1.

Bill Yuchen Lin, Ronan Le Bras, Kyle Richardson, Ashish Sabharwal, Radha Poovendran, Peter Clark, et al. Zebralogic: On the scaling limits of llms for logical reasoning, 2025. https://arxiv.org/abs/2502.01100. Cited at: 1.

Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. Is your code generated by chatgpt really correct? rigorous evaluation of large language models for code generation, 2023. https://arxiv.org/abs/2305.01210. Cited at: 1.

Jongho Park, Vasilis Kontonis, Shivam Garg, Akshay Krishnamurthy, and Dimitris Papailiopoulos. Scaling discovery through test-time communication, 2026. https://arxiv.org/abs/2609.21032. Cited at: 1.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters, 2024. https://arxiv.org/abs/2408.03314. Cited at: 1.

Zafir Stojanovski, Oliver Stanley, Joe Sharratt, Richard Jones, Abdulhakeem Adefioye, Jean Kaddour, et al. Reasoning gym: Reasoning environments for reinforcement learning with verifiable rewards, 2025. https://arxiv.org/abs/2505.24760. Cited at: 1.

Leo Tolstoy. *Anna Karenina*. Penguin Classics, 2014. ISBN 9780141199610. https://www.penguinrandomhouse.com/books/530530/anna-karenina-by-leo-tolstoy/9780141199610/. Translated by Richard Pevear and Larissa Volokhonsky; Part One, Chapter 1. Cited at: 1.

Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. Planbench: An extensible benchmark for evaluating large language models on planning and reasoning about change, 2023. https://arxiv.org/abs/2206.10498. Cited at: 1.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, et al. Self-consistency improves chain of thought reasoning in language models, 2023. https://arxiv.org/abs/2203.11171. Cited at: 1.

Terry Yue Zhuo, Minh Chien Vu, Jenny Chim, Han Hu, Wenhao Yu, Ratnadira Widyasari, et al. Bigcodebench: Benchmarking code generation with diverse function calls and complex instructions, 2025. https://arxiv.org/abs/2406.15877. Cited at: 1.

A Study Inventory and Protocols

This appendix documents the eleven complementary studies represented in the normalized evidence export. It omits prospective scaling models, unfinished experiments, and imported full-benchmark baselines. Study codes below identify the complete cell table in the versioned repository and the machine-readable evidence bundle. Counts are completed scientific trial keys, not request attempts.

Code	Study	Cal. tasks	Cal. trials	Eval. tasks	Eval. trials
BC-L	BigCodeBench Luna 5.6	32	1,792	32	1,728
BC-M	BigCodeBench Muse	32	1,792	32	1,728
MP-L	MBPP+ Luna 6	16	448	16	432
MP-M	MBPP+ Muse	16	448	16	432
Z-O	ZebraLogic original	32	1,792	64	3,456
Z-C	ZebraLogic confirmation	0	0	300	6,600
GC-L	Graph colouring Luna 6	0	0	100	1,200
GC-M	Graph colouring Muse	0	0	100	1,200
CD	Countdown Luna 6	0	0	100	1,200
BB	BBEH subset Luna 6	0	0	100	1,200
PB	PlanBench subset Luna 6	0	0	100	1,200
	Total		6,272		20,376

Table A1 | Scientific trial inventory. Every non-MBPP+ cell has two repetitions per task; MBPP+ has one. Calibration and evaluation tasks are distinct within the original studies. The two graph-colouring studies reuse the same tasks; cross-model task counts must not be added as independent samples. Separate pilots and smoke checks are not calibration trials in this inventory.

A.1 Design and selection

The five original grids evaluate widths 1, 2, 4, 8, and 16 at three evaluation ceilings, with distinct calibration tasks and shared single-worker controls. Draft, revision, and model integration receive nominal shares of 50%, 25%, and 25%. The five smaller diagnostic studies compare independent one, independent four, and ring four at two ceilings, using draft and revision shares of 50% each without a model integrator. Ring revision receives a predecessor draft; independent revision receives the worker's own draft. Prompt costs and output reserves can prevent calls from launching.

BC-L records `gpt-5.6-luna`; later Luna studies request `openai/gpt-6-luna`; Muse records `meta/muse-spark-1.3-contributor`. Models, routing, and sampling settings vary by study. Recorded activation is available only for the four programming studies. Exact grids, recorded configurations, task-source revisions, excerpt limits, and available dependency records are in the repository's Methods document and configuration export.

A.2 Task construction and confirmation

Graph colouring uses 100 fixed, locally generated greedy-solvable three-colour graphs: 20 each at 10, 12, 14, 16, and 18 vertices, with edge probability 0.2. Both models share these tasks, giving 1,200 evaluation trials per model. This is a local adaptation inspired by Reasoning Gym, not its official task distribution. Countdown uses 100 algorithmically generated puzzles with four to eight supplied numbers. Both families have deterministic correctness checks; valid solutions need not be unique. BBEH uses 25 tasks from each of four categories; PlanBench uses 100 Blocksworld Hard instances and a local STRIPS interpreter. These subsets are not full-benchmark or unaided-collaboration evaluations.

Original ZebraLogic compares calibration-selected width policies with its selected fixed-width baseline. Confirmation transfers communicating widths 4/8/16 without refitting to 300 unused puzzles, with 12 per size stratum. Its primary comparison uses communicating fixed

Studies	Final candidate construction	Final task criterion
BigCodeBench	Model integration of worker revisions	Saved sandbox test outcome; no hidden-test feedback to workers.
MBPP+	Model integration of worker revisions	Base and EvalPlus tests.
ZebraLogic	Model integration of worker revisions	Normalized whole-grid exact correctness; malformed/empty grids fail.
Graph colouring	First locally valid revision in worker order; otherwise first nonempty revision; no draft rescue	Local three-colour validity for the generated graph.
Countdown	Verifier-assisted selection of a valid candidate	Local arithmetic constraints and target match.
BBEH subset	Gold-blind plurality of normalized answers; worker-order tie breaking	Local normalized exact/fuzzy matching to the reference answer.
PlanBench subset	First locally valid revision in worker order	Local STRIPS plan validity; optimality is not required.

Table A2 | Selection and grading are distinct parts of the evaluated system. Deterministic local grading is not independent validation of grader correctness or benchmark representativeness.

four at the two higher ceilings. Shared cells are deduplicated into eleven configurations and 6,600 trials. The complete grids and generator settings are in the repository.

B Resource Accounting, Failures, and Provenance

Retained tokens combine reported usage, local estimates, and conservative full-allocation debits for eligible unknown-usage calls, depending on the study. They are not exact physical compute or complete attempted-call expenditure. Comparisons use the same study and ceiling.

Twelve BC-M calibration trials exceed their nominal ceilings by at most 1.92%. No retained evaluation ledger total exceeds its nominal ceiling; that check does not establish that every provider call or unobserved attempt respected its allocation. Seven Muse graph-colouring trials at 8,192 were classified as budget-violation failures under a post-start amendment. They remain operational zeros, not direct evidence of reasoning errors. Actual total tokens remain unknown for both graph models, as does Muse's total cost. Attempt totals, known costs, missing-usage counts, and the seven failure keys are preserved in the repository's Data document and graph export.

PlanBench's independent four-worker cells retain 1,600 stage calls per ceiling: 1,599 finish at the length limit at 8,192 and three at 16,384. The single-worker cells retain 400 calls each without length-limit finishes. These observations diagnose feasibility but do not isolate a causal effect of truncation.

Amendments and exclusions. Original ZebraLogic and confirmation include post-start accounting or retry amendments and cannot be described as uniformly preregistered. Pilots, smoke checks, imported baselines, incomplete studies, and one superseded MBPP+ Muse row are excluded. Pilot truncation warnings were acknowledged before some frozen evaluations continued. Detailed operational records and excluded-check counts are documented in the repository; retained outcomes do not imply zero expenditure on failed attempts.

Analysis artifacts and availability. The versioned public repository provides all 26,648 normalized retained trials, all 316 cell summaries, a readable table of all 176 evaluation cells, extended protocols, recorded configurations, provenance hashes, and the offline aggregate

verifier.[2] The verifier checks saved aggregates and point estimates; it does not recompute intervals, validate graders, or replay inference. The shared graph export includes task edges, witness colourings, and operational-failure keys. The controlled-study note contains the supplied protocol and result summary and states which supporting records are available. Its coverage is separate from the eleven complementary studies checked by the offline verifier. Full experimental source/run archives, historical reports, and raw provider payloads are outside this concise public package.

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C Paired Inference and Sensitivity Analyses

Tasks are the resampling clusters, retaining all their configurations and repetitions so contrasts remain paired. Confirmation uses size strata; diagnostic studies use graph size, Countdown size, BBEH category, or PlanBench object count. Intervals are conditional and study-specific, with no pooled meta-analysis or cross-study multiplicity correction. Resampling draw counts, seeds, and manifest-order conventions are recorded in the repository's Methods document and configuration export.

Study	Ring minus independent (pp)	95% interval
BigCodeBench Luna 5.6	$+0.39$	$[-1.04, 2.34]$
BigCodeBench Muse	-1.43	$[-3.65, 0.39]$
MBPP+ Luna 6	-1.56	$[-4.17, 0.00]$
MBPP+ Muse	$+2.60$	$[0.00, 7.81]$
Graph colouring Luna 6	-2.00	$[-4.00, -0.25]$
Graph colouring Muse	-0.75	$[-2.25, 0.75]$
Countdown	-5.00	$[-8.50, -1.75]$
BBEH subset	$+1.75$	$[-0.50, 4.00]$
PlanBench subset[†]	$+0.25$	$[0.25, 0.25]$

Table C1 | Every saved within-study ring contrast, averaging the evaluated multi-worker widths, ceilings, and repetitions. ZebraLogic has distinct selected-policy contrasts, described below. [†]The PlanBench interval reflects a singleton stratum, not population-level certainty.

Sparse-stratum sensitivity. PlanBench's ring advantage comes from one passing repetition on the sole task in its seven-object stratum. Stratified resampling always retains that task, producing the degenerate $[0.25, 0.25]$-pp interval. An unstratified paired-task sensitivity interval is $[0.00, 0.75]$ pp. This does not establish general communication superiority or zero population uncertainty.

Policy comparator sensitivity. The original ZebraLogic primary contrast is $+3.776$ pp, interval $[1.302, 6.510]$, against the calibration-selected fixed-width baseline, averaging conditions and ceilings. Confirmation gives $+0.917$ pp, interval $[-0.583, 2.500]$, against communicating fixed four at the two higher ceilings. These are different estimands and must not be pooled. An interval spanning zero does not establish equivalence. Full configuration counts are provided in the repository's evaluation-cell table.

[2] https://github.com/PluvioXO/agent-swarms-reproducibility/tree/reproducibility-v1.3